

Learning to Listen:

Scalable Bird Species Detection from Environmental Audio Using Deep Convolutional Networks

Saurav Bellam

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1 Introduction

Ecosystems across the globe are facing unprecedented pressure, and effective conservation increasingly depends on our ability to monitor biodiversity at scale. In many of the world’s most remote and ecologically rich environments, direct observation is limited, making it difficult to assess species presence and ecosystem health. In this context, passive acoustic monitoring has emerged as a powerful alternative—allowing researchers to “listen” to ecosystems and capture the vocal activity of wildlife over extended periods. Building on this paradigm, our research leverages deep learning, specifically Convolutional Neural Networks (CNNs), to develop a model capable of automatically identifying species from continuous environmental audio recordings.

The focus of this study is the Pantanal wetlands of Brazil, one of the largest and most biodiverse ecosystems on Earth, spanning over 150,000 km² and home to more than 650 bird species alongside countless other taxa. Despite its ecological importance, much of the Pantanal remains under-monitored due to its vast scale, seasonal flooding, and increasing threats from wildfires, agricultural expansion, and climate change. To address these challenges, a growing network of acoustic sensors continuously records soundscapes across the region, producing massive volumes of audio data that are infeasible to analyze manually. This creates a critical need for automated, scalable methods capable of extracting meaningful ecological information from noisy, real-world recordings.

In this work, we explore the application of deep transfer learning to passive acoustic monitoring, using a dataset composed of field recordings sourced from platforms such as iNaturalist and Xeno-canto. These datasets provide diverse but weakly labeled audio samples, reflecting real-world conditions where multiple species may be present simultaneously and annotations are often incomplete. By training CNN-based models on this data, we aim to develop a robust classification framework capable of identifying understudied species within complex soundscapes.

1.1 Problem Statement

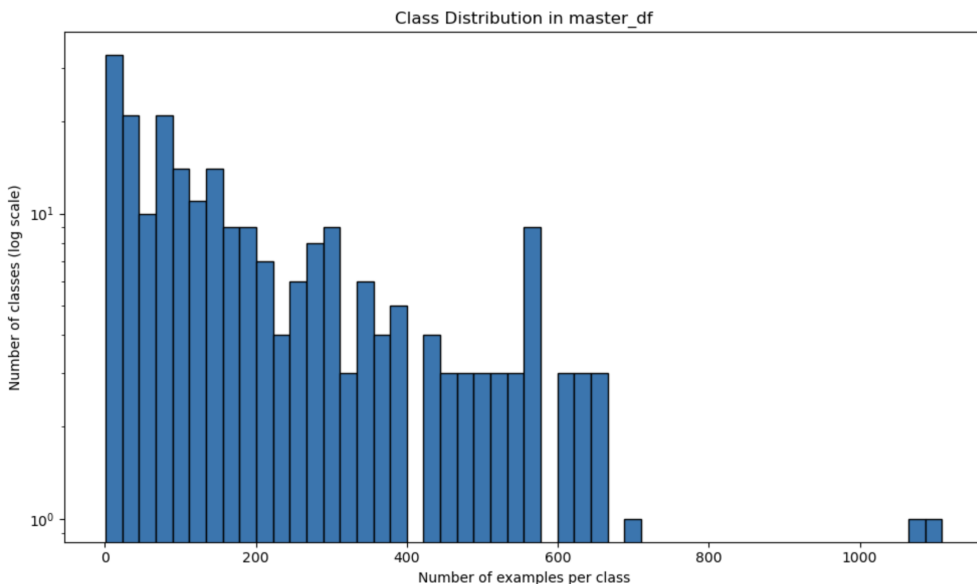


Figure 1: Log-scaled histogram of per-class sample counts in the BirdCLEF dataset, illustrating a strongly long-tailed distribution. A small number of head classes contain the majority of samples, while most classes (tail species) are sparsely represented.

As shown in Figure 1, the BirdCLEF dataset exhibits a strongly long-tailed distribution, where a small number of classes dominate the sample count while the majority remain underrepresented.

This dataset contains a heavy class imbalance, with a majority of the heavily represented classes belonging to the avian taxonomy. This bias in the dataset would cause severe overfitting, with the model disproportionately learning dominant bird species while failing to generalize to underrepresented or rare classes. As a result, predictions become skewed toward head classes, leading to poor performance on tail species and limiting the

model’s effectiveness in real-world biodiversity monitoring scenarios.

2 Methods

2.1 Data and Preprocessing

We utilize the BirdCLEF dataset, consisting of standard labeled recordings, continuous soundscape recordings, and taxonomy metadata. The task is formulated as multilabel classification at the recording level.

Audio preprocessing follows a standardized pipeline. All recordings are converted to mono by averaging channels and resampled to 32,000 Hz. Each recording is segmented into non-overlapping 5-second clips:

$$X = \{x_1, x_2, \dots, x_T\}$$

with zero-padding applied to the final segment if necessary.

Each clip is converted into a 3-channel spectrogram stack consisting of log-mel, delta, and delta-delta features. Spectrograms are computed using:

$$n_{\text{fft}} = 2048, \quad \text{hop_length} = 512, \quad n_{\text{mels}} = 128$$

with frequency bounds:

$$f_{\min} = 300 \text{ Hz}, \quad f_{\max} = 16000 \text{ Hz}$$

and resized to a fixed resolution of 128×384 .

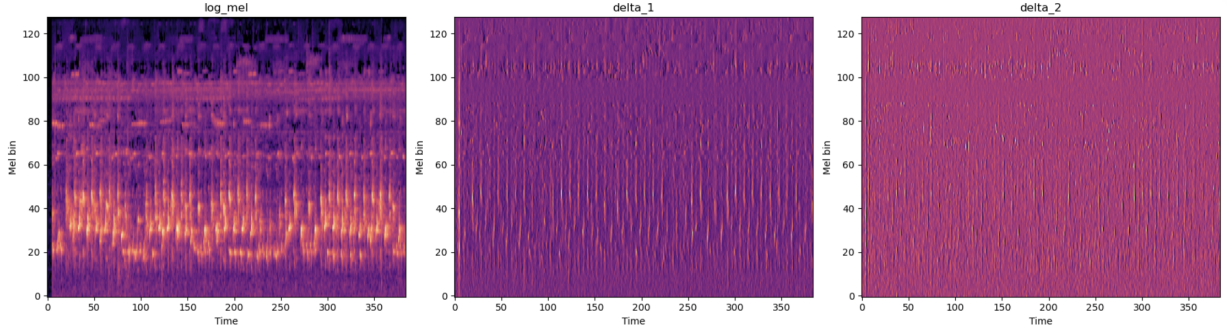


Figure 2: Example input representation consisting of log-mel, delta, and delta-delta spectrogram channels.

An example of the resulting spectrogram representation is shown in Figure 2.

To reduce amplitude-scale variation, each spectrogram channel is normalized independently per frequency bin across time:

$$\mu_f = \frac{1}{T} \sum_{t=1}^T S_{f,t}, \quad \sigma_f = \sqrt{\frac{1}{T} \sum_{t=1}^T (S_{f,t} - \mu_f)^2 + \epsilon}$$

$$\tilde{S}_{f,t} = \frac{S_{f,t} - \mu_f}{\sigma_f}$$

Recordings are grouped into quality subsets: clean (rating = 5), semi ($2.5 \leq \text{rating} < 5$), noisy (rating < 2.5), and soundscape.

2.2 Model Architecture

Raw audio	→	Mono + 32 kHz	→	5-second clips	→	3-channel spectrogram
Log-mel, delta, delta-delta	→	Per-frequency normalization	→	ResNet-style CNN	→	Clip embeddings h_t
Class-wise attention	→	Weighted aggregation z_c	→	Linear logits $\hat{y}_c$	→	Sigmoid probabilities p_c

Figure 3: Full model pipeline. Raw recordings are converted to mono, resampled, segmented into 5-second clips, transformed into 3-channel spectrogram stacks, normalized, encoded using a ResNet-style CNN, and aggregated using class-wise attention pooling to produce multilabel species predictions.

Each clip is encoded independently using a Convolutional Neural Network (CNN) based on a ResNet-style architecture. Given an input spectrogram clip $x_t \in \mathbb{R}^{3 \times H \times W}$, where the three channels correspond to log-mel, delta, and delta-delta features, the network extracts a fixed-dimensional embedding:

$$h_t = f_\theta(x_t), \quad h_t \in \mathbb{R}^d$$

with embedding dimension $d = 96$.

The use of three-channel input allows the model to leverage temporal dynamics in addition to static spectral information. The log-mel channel captures frequency energy distribution, while the delta and delta-delta channels encode first- and second-order temporal derivatives. This stacked representation aligns naturally with standard CNN architectures designed for RGB images, and has been shown to improve performance by enabling the network to learn joint spectro-temporal patterns.

The encoder follows a ResNet-style design consisting of an initial convolutional stem followed by a sequence of residual blocks. The stem applies a 7×7 convolution with stride 2, followed by batch normalization, a ReLU activation, and a 3×3 max pooling layer. This is followed by multiple stages of residual blocks, each containing two 3×3 convolutional layers with batch normalization and ReLU activations. Skip connections are used to add the input of each block to its output, enabling stable gradient flow and improved training of deeper networks.

Spatial downsampling is performed via strided convolutions at the beginning of each stage, progressively reducing the spatial resolution while increasing the number of feature channels. The base channel width is set to 24, and channel depth increases across stages in a standard ResNet progression.

After the final convolutional stage, global average pooling is applied across the spatial dimensions to produce a compact feature vector:

$$h_t = \text{GAP}(F(x_t))$$

where $F(\cdot)$ represents the convolutional backbone. A dropout layer with rate 0.2 is applied to the resulting embedding to reduce overfitting.

These embeddings are then aggregated using class-wise attention pooling. For each class c , attention scores are computed:

$$e_{t,c} = g_c(h_t)$$

and normalized across clips:

$$\alpha_{t,c} = \frac{\exp(e_{t,c})}{\sum_{t'=1}^T \exp(e_{t',c})}$$

The class-specific representation is computed as:

$$z_c = \sum_{t=1}^T \alpha_{t,c} h_t$$

Final logits are produced via a linear classifier:

$$\hat{y}_c = w_c^\top z_c + b_c$$

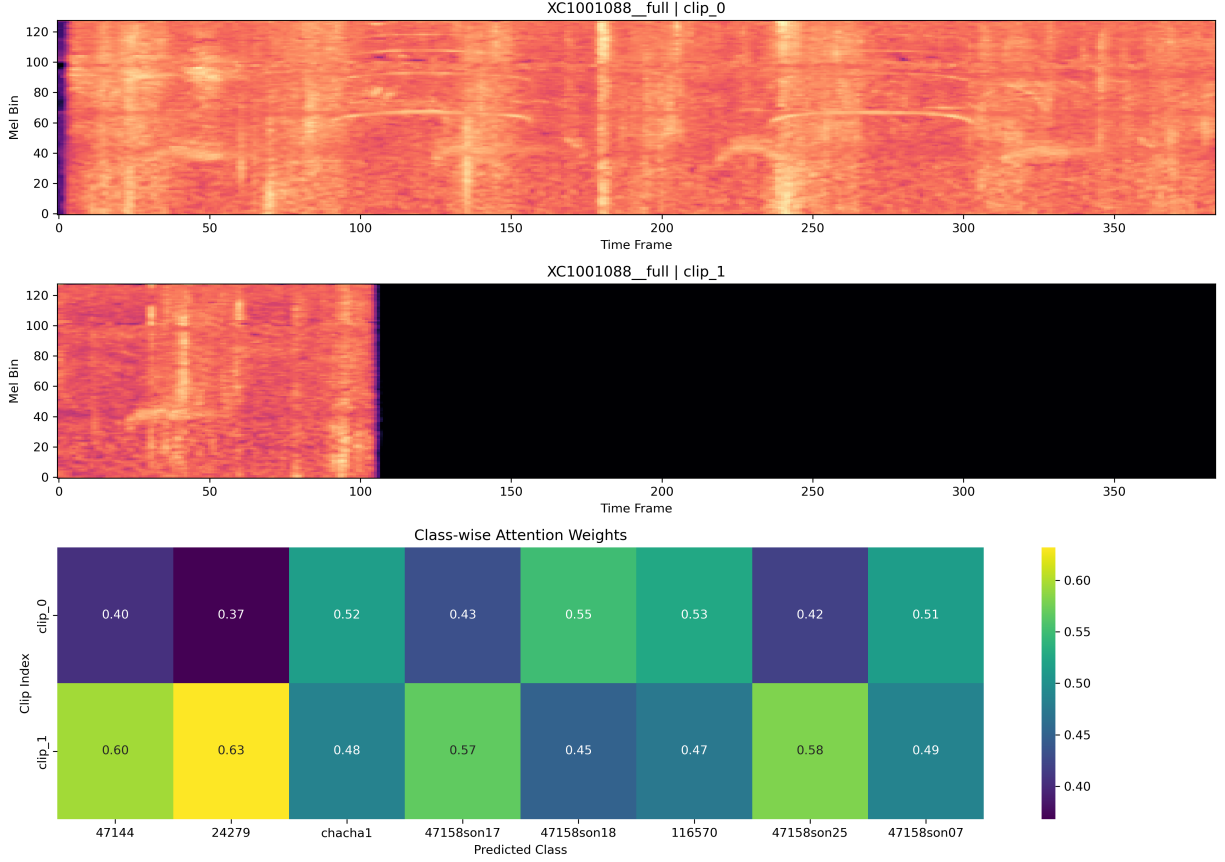


Figure 4: Example visualization of class-wise attention pooling. The top panels show two sequential 5-second spectrogram clips from the same recording, while the bottom heatmap shows the learned attention weights for each predicted class across clips. Brighter values indicate that the model assigns more importance to that clip for a given class.

with probabilities given by:

$$p_c = \frac{1}{1 + \exp(-\hat{y}_c)}$$

Class-wise attention pooling is used to aggregate multiple 5-second clips into a single recording-level multi-label prediction. Since each recording may contain several clips, not every clip is equally useful for predicting every species. Some clips may contain the target vocalization, while others may contain silence, background noise, overlapping species, or unrelated environmental sounds. Instead of treating all clips equally with mean pooling, class-wise attention allows the model to learn which clips are most informative for each species separately.

For each clip t , the CNN encoder produces a clip-level embedding h_t . Then, for each class c , a small attention network computes an attention score:

$$e_{t,c} = g_c(h_t)$$

where $e_{t,c}$ represents how relevant clip t is for predicting class c . These scores are normalized across all clips in the recording using a softmax function:

$$\alpha_{t,c} = \frac{\exp(e_{t,c})}{\sum_{t'=1}^T \exp(e_{t',c})}$$

Here, $\alpha_{t,c}$ is the attention weight assigned to clip t for class c , and T is the total number of clips in the recording. Because the softmax is applied across clips for each class, the attention weights for a given class sum to one:

$$\sum_{t=1}^T \alpha_{t,c} = 1$$

The final class-specific recording representation is then computed as a weighted sum of clip embeddings:

$$z_c = \sum_{t=1}^T \alpha_{t,c} h_t$$

This means that each class receives its own pooled representation z_c , rather than forcing all species to share the same pooled recording embedding. This is important for multilabel bird sound classification because different species may occur at different moments within the same recording.

Conceptually, class-wise attention pooling acts similarly to a weak event detector. The model is only given recording-level labels, not exact timestamps for when each species vocalizes. However, attention pooling allows the model to assign higher weights to clips that contain acoustic patterns useful for a particular species and lower weights to clips that mostly contain background noise or irrelevant sounds. In this way, the model can partially mitigate weak labeling by learning which temporal regions are most relevant to each label.

Figure 4 illustrates this behavior. The first two panels show the spectrograms for two clips from the same recording. The bottom heatmap shows the learned attention weights for several predicted classes. For some classes, the model places more attention on `clip_0`, while for others it places more attention on `clip_1`. This demonstrates why class-wise attention is useful: different species predictions can depend on different parts of the recording, rather than forcing one global pooling decision for all classes.

Final logits:

$$\hat{y}_c = w_c^\top z_c + b_c$$

$$p_c = \frac{1}{1 + \exp(-\hat{y}_c)}$$

2.3 Training Procedure

Training follows a three-phase curriculum:

Phase 1: Family-balanced sampling with target 20 examples per family. Data is selected in order: clean $\rightarrow$ semi $\rightarrow$ noisy $\rightarrow$ soundscape (if needed). Loss: unweighted BCE. Learning rate: 10^{-3} .

Phase 2: Clean + semi + noisy data using class-aware sampling with weights proportional to:

$$w_c \propto \frac{1}{\sqrt{\text{class_count}_c}}$$

Loss: unweighted BCE. Learning rate: 5×10^{-4} .

Phase 3: Full dataset including soundscapes using natural shuffled sampling. Loss: class-weighted BCE:

$$L = \frac{1}{C} \sum_{c=1}^C [-\beta_c y_c \log(p_c) - (1 - y_c) \log(1 - p_c)]$$

Learning rate: 10^{-4} .

Optimization is performed using AdamW with weight decay 10^{-4} and no data augmentation.

2.4 Validation and Model Selection

A 10% internal validation split is used for model selection and plateau detection, stratified by soundscape presence when possible. Training in each phase continues until validation macro ROC-AUC fails to improve by at least 0.001 for 6 consecutive epochs. The best checkpoint across all phases is retained.

2.5 Evaluation Metrics

Performance is evaluated using macro ROC-AUC and macro PR-AUC. Additional evaluations include per-class ROC-AUC, rare-class ROC-AUC, and family-level ROC-AUC (computed via max pooling over species within a family).

Classes are grouped as:

- head / medium / tail: frequency tertiles over training data
- rare: low-frequency classes
- missing-from-clean: absent in clean subset
- soundscape-heavy: $\geq 50\%$ soundscape samples

2.6 Reproducibility

Audio was converted to mono and resampled to 32 kHz. Recordings were segmented into non-overlapping 5-second clips. Each clip was represented as a 3-channel spectrogram stack using $n_{\text{fft}} = 2048$, $\text{hop_length} = 512$, $n_{\text{mels}} = 128$, $f_{\text{min}} = 300$ Hz, and $f_{\text{max}} = 16000$ Hz, resized to 128×384 . Each channel was normalized per frequency bin across time. The final model used a CNN encoder with base width 24, embedding dimension 96, dropout 0.2, AdamW optimization, batch size 1, and no data augmentation.

3 Results

3.1 Training Setup

All results reported in this work are based on a single final training run rather than cross-validation. An internal validation split comprising 10% of the training data was used exclusively for model selection and phase-wise stopping criteria. No data augmentation was applied during this final run in order to isolate the effects of the curriculum learning strategy and model architecture.

The model was trained using the AdamW optimizer with a batch size of 1. The convolutional encoder was configured with a base channel width of 24, an embedding dimension of 96, and a dropout rate of 0.2. During training, the number of clips per recording was capped at a maximum of 8 clips per sequence to maintain computational efficiency and consistent input structure.

Training followed a three-phase curriculum learning strategy. In the first phase, the model was trained on a family-balanced subset constructed using a clean-first selection strategy, prioritizing high-quality examples while maintaining taxonomic diversity. In the second phase, the training data was expanded to include clean, semi, and noisy recordings, allowing the model to learn broader species-level variation. In the final phase, soundscape recordings were incorporated alongside all previous data, aligning the training distribution with real-world deployment conditions.

Each training phase continued until validation macro ROC-AUC failed to improve by at least 0.001 for 4 consecutive epochs, corresponding to a patience value of 4. Model selection was performed by retaining the checkpoint that achieved the highest validation macro ROC-AUC across all phases.

3.2 Main Results

The final class-wise attention model achieved a validation macro ROC-AUC of 0.5982 and a holdout test macro ROC-AUC of 0.4770. Corresponding macro PR-AUC values were 0.0388 on validation and 0.2544 on the holdout test set. Rare-class performance followed a similar pattern, with validation rare-class ROC-AUC of 0.6236 decreasing to 0.4661 on the holdout set. Family-level performance was stronger on the internal validation split, with 0.6987 macro ROC-AUC, than on holdout test data, with 0.4248 macro ROC-AUC. As shown in Figure 5, the model demonstrates convergence during training, although this does not fully transfer to holdout performance.

A substantial drop in performance is observed when transitioning from validation to holdout test data. Macro ROC-AUC decreases from 0.5982 to 0.4770, rare-class ROC-AUC drops from 0.6236 to 0.4661, and family-level macro ROC-AUC declines from 0.6987 to 0.4248. This indicates that the model learns meaningful structure on

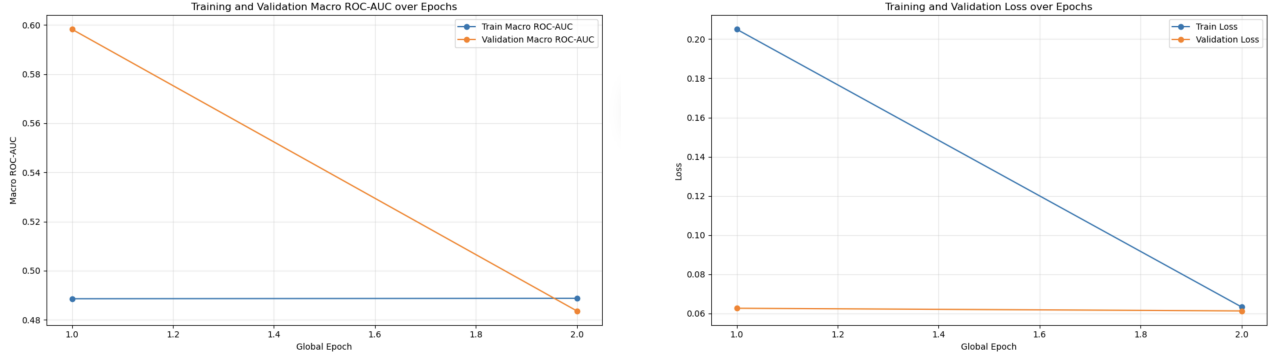


Figure 5: Training dynamics across epochs showing ROC-AUC and loss behavior.

Table 1: Overall performance on validation and holdout test sets.

Metric	Validation	Holdout Test
Macro ROC-AUC	0.5982	0.4770
Macro PR-AUC	0.0388	0.2544
Rare-class ROC-AUC	0.6236	0.4661
Family-level ROC-AUC	0.6987	0.4248

the internal validation split, but generalization to the more challenging soundscape-style holdout data remains limited.

3.3 Subgroup Analysis

Performance varied substantially across class-frequency groups. On the holdout set, medium-frequency classes achieved the strongest macro ROC-AUC of 0.6935, while head classes were notably weaker at 0.3671. Tail classes reached 0.4661 macro ROC-AUC, and missing-from-clean classes achieved 0.4678. Soundscape-heavy classes obtained 0.4990 macro ROC-AUC on holdout data, indicating partial but limited transfer to deployment-like conditions.

Table 2: Subgroup performance measured by macro ROC-AUC.

Class Group	Validation	Holdout Test
Head	0.5850	0.3671
Medium	0.6174	0.6935
Tail	0.6091	0.4661
Missing-from-clean	0.6407	0.4678
Soundscape-heavy	0.7333	0.4990

Interestingly, medium-frequency classes outperform both head and tail classes on the holdout set, while head classes exhibit unexpectedly weak performance. Soundscape-heavy classes show moderate performance but remain below robust levels, suggesting incomplete adaptation to deployment conditions.

3.4 Family-Level Results

Family-level analysis reveals that the model captures higher-level acoustic structure more effectively than fine-grained species distinctions. On the validation set, reptilia, insecta, and amphibia exhibit strong performance, while aves and mammalia are weaker. On the holdout test set, reptilia remains relatively strong, but performance for other families degrades significantly, particularly mammalia.

These results further reinforce the gap between validation and holdout performance, demonstrating that family-level discrimination learned during training does not consistently transfer to unseen environments.

Table 3: Family-level ROC-AUC.

Family	Validation	Holdout Test
Reptilia	0.9494	0.7642
Insecta	0.8725	0.5081
Amphibia	0.8054	0.3333
Mammalia	0.5185	0.0243
Aves	0.3476	0.4942

3.5 Training Dynamics

Training dynamics indicate that the best-performing model checkpoint was obtained during Phase 3 of the curriculum learning procedure. The final validation macro ROC-AUC reached 0.6783 prior to plateau-based stopping, which occurred after 24 epochs in the final phase.

Although Phase 3 improved validation performance, these gains did not fully translate to holdout test performance. This suggests that while the curriculum learning strategy enhances the model’s ability to fit training and validation distributions, it is insufficient on its own to achieve robust generalization to real-world soundscape data.

4 Discussion

4.1 Generalization Gap

A central finding of this work is the substantial performance gap between validation and holdout test results. While the model achieved a validation macro ROC-AUC of 0.5982, performance dropped to 0.4770 on the holdout test set. Similar declines were observed across rare-class and family-level metrics, indicating that the learned representations did not generalize robustly to unseen soundscape data.

This gap suggests that the model is able to capture meaningful structure within the training and validation distributions but struggles to transfer this knowledge to more realistic and noisy deployment conditions. The holdout test set, which more closely resembles continuous environmental recordings, introduces additional variability that is not fully captured during training, particularly in terms of background noise, overlapping species, and recording artifacts.

4.2 Impact of Weak Labels and Noise

One key factor contributing to this generalization failure is the weakly labeled nature of the dataset. Labels are provided at the recording level without precise temporal annotations, meaning that many clips may not actually contain the labeled species. As a result, the model is trained with noisy supervision, where incorrect or ambiguous labels introduce uncertainty into the learning process.

This issue is amplified in soundscape recordings, where multiple species may overlap and background noise is prevalent. In such settings, the model must both detect the presence of species and localize relevant acoustic events without explicit guidance. Although the class-wise attention mechanism partially addresses this by allowing the model to focus on informative clips, it is still limited by the quality of the supervision signal.

4.3 Rare-Class Performance

Despite these challenges, an encouraging result is the relatively strong performance on rare classes in the validation setting. The validation rare-class ROC-AUC of 0.6236 exceeds the overall macro ROC-AUC, indicating that the model is capable of learning discriminative features for underrepresented species. This suggests that the curriculum learning strategy and class-aware sampling were effective in mitigating some of the effects of class imbalance.

However, this advantage does not fully transfer to the holdout test set, where rare-class performance drops to 0.4661. While this decline mirrors the overall generalization gap, the ability of the model to perform competitively on rare classes demonstrates that one of the primary goals of the work, improving representation of tail species, has been partially achieved.

4.4 Subgroup and Family-Level Behavior

Analysis across class-frequency groups reveals several important patterns. On the holdout set, medium-frequency classes outperform both head and tail classes, suggesting that they provide a balance between sufficient training data and reduced overfitting. In contrast, head classes exhibit unexpectedly poor performance, likely due to over-reliance on dataset-specific patterns that do not generalize well.

Family-level results further highlight the model’s strengths and limitations. High validation performance for groups such as reptilia and insecta indicates that the model can learn coarse-grained acoustic structure. However, the sharp decline in holdout performance, particularly for mammalia, suggests that these representations are not robust across environments.

4.5 Limitations and Future Work

Several limitations of the current approach contribute to the observed performance gap. First, the absence of data augmentation reduces the model’s exposure to variability in acoustic conditions, limiting its ability to generalize. Second, weak labeling introduces unavoidable noise into the training process, which cannot be fully resolved by attention mechanisms alone. Third, the use of a fixed clip-based representation may fail to capture longer temporal dependencies present in continuous recordings.

Future work should focus on improving robustness to real-world conditions. Incorporating audio data augmentation techniques such as time masking, frequency masking, and background noise injection could improve generalization. Additionally, semi-supervised or pseudo-labeling approaches may help refine weak labels and reduce supervision noise. More advanced temporal modeling, such as sequence-level transformers or event detection frameworks, could further improve the model’s ability to localize and identify species within complex soundscapes.

Finally, domain adaptation techniques may help bridge the gap between curated training data and deployment environments. Explicitly modeling the differences between clean recordings and soundscapes could enable more consistent performance across diverse ecological conditions.

4.6 Summary

Overall, the proposed model demonstrates the ability to learn meaningful acoustic representations and partially address class imbalance, particularly for rare species. However, significant challenges remain in achieving robust generalization to real-world soundscape data. These findings highlight the importance of addressing weak supervision, noise, and domain shift in future work on large-scale bioacoustic classification.

5 Conclusion

In this work, we developed a CNN-based bioacoustic classification framework with class-wise attention pooling to identify species from continuous environmental audio recordings. The proposed pipeline integrates a ResNet-style encoder with a curriculum learning strategy designed to address the challenges of weak labels, noisy recordings, and severe class imbalance inherent in the BirdCLEF dataset.

Experimental results demonstrate that the model is capable of learning meaningful acoustic representations, achieving a validation macro ROC-AUC of 0.5982. Notably, performance on rare classes was relatively strong in the validation setting, indicating that the curriculum learning approach and class-aware sampling were effective in improving representation of underrepresented species. This suggests that one of the primary objectives of the work—enhancing performance on tail classes—was partially achieved.

However, a significant drop in performance was observed when evaluating on the holdout test set, with macro ROC-AUC decreasing to 0.4770. This highlights a key limitation of the current approach: insufficient generalization to real-world soundscape data. The results suggest that while the model captures structure within the training distribution, it struggles to transfer this knowledge to more complex and noisy deployment environments.

Further analysis revealed that medium-frequency classes performed best on the holdout set, while head classes underperformed, likely due to overfitting to dataset-specific patterns. Additionally, family-level results indicate that the model more reliably captures coarse-grained acoustic structure than fine-grained species-level

distinctions. These findings reinforce the challenges posed by weak supervision, domain shift, and environmental variability in large-scale bioacoustic classification tasks.

Future work should focus on improving robustness and generalization through techniques such as audio data augmentation, semi-supervised learning, and pseudo-labeling to better handle weak annotations. Incorporating more advanced temporal modeling methods and domain adaptation strategies may further enhance performance in real-world soundscape conditions.

Overall, this study demonstrates the potential of combining deep convolutional architectures with attention-based pooling and curriculum learning to tackle challenging bioacoustic classification problems, while also highlighting the critical need for methods that can effectively bridge the gap between curated training data and real-world deployment environments.